Machine Learning Applications in Option Pricing, Implied Volatility, and Bid-Ask Spread Prediction

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Supplementary materials:

Google Colab Notebook

1 Introduction

Option pricing is one of the central problems in financial economics because option values affect trading, hedging, portfolio insurance, and risk management. Classical models, the most famous being Black-Scholes, offer a beautiful analytical framework, which is however not always consistent with observed market behaviour due to its simplifying assumptions. In particular, constant volatility assumptions struggle to explain empirical patterns such as the volatility smile or skew, while American-style exercise features create additional complexity because valuation becomes an optimal stopping problem rather than a straightforward closed-form calculation. These shortcomings inspired the construction of stochastic-volatility models like Hull and White [1], and more efficient approximation and simulation methods of American options.

At the same time, modern derivatives markets generate large amounts of structured data, making machine learning a natural complementary approach. Instead of introducing a strict parametric framework, machine learning models are able to learn nonlinear relationships that are complex to establish based on data. Previous studies demonstrate that both non-parametric and machine learning techniques are good at pricing options and can outperform classical benchmark models when market conditions do not adhere to idealized assumptions. This benefit is particularly applicable in environments where the price-setting process is very nonlinear, or the influence of transaction costs and market microstructure frictions on quoted prices [2].

In this report, machine learning is applied to three related tasks in the option-market. First, a synthetic dataset of American call options is taken to test the extent to which various machine learning models can replicate a pricing function created in a controlled setting. Second, implied volatility is predicted based on a real-world NFLX options dataset to reflect the forward looking evaluation of uncertainty by the market. Third, the bid-ask spreads are modelled by using the same real-world data and they give information on the liquidity, the cost of trading and the risk compensation of the market-makers. Combined, these three tasks enable machine learning to be tested as a pricing instrument, as well as a means to explain significant cross-sectional behaviors in actual derivatives markets. The value of this work is thus two in one. It can be methodologically compared to a large number of regression models as linear regression, support vectors regression, decision trees, random forest, gradient boosting, XGBoost, LightGBM, and neural networks are all compared using a uniform evaluation structure. On the financial front, it connects the empirical findings to the option-pricing theory, implied-volatility surface behaviour, and market microstructure intuition. Moreover, to analyze whether the most significant predictors that machine learning finds have economic justification, model interpretability tools are employed.

The rest of the report is presented in the following way. Section 2 examines the literature on classical option pricing and machine learning techniques, implied volatility, and the formation of bid-ask spreads. Section 3 outlines the datasets, cleaning procedure, and finance-inspired feature engineering. Section 4 defines modelling framework and evaluation measures. Empirical results, discussion, limitations and

conclusions are provided later.

2 Background and Literature Review

2.1 Classical option pricing and the move beyond Black-Scholes

The Black-Scholes model has been the mainstay of the current option pricing, but its limitations in practice are familiar. Specifically, the volatility is assumed to be constant, which is not consistent with the empirical data that the implied volatility is usually different across the strike prices and maturities. Hull and White demonstrated that in cases where the volatility is a stochastic process, Black-Scholes can systematically misprice options, particularly around the money, and that the amount of underpricing will tend to rise with time to maturity. Their findings assisted in providing a stochastic volatility as an extension to the classical model [1] that is essential.

For American options, the issue is more challenging since the holder can exercise before maturity. This implies that the valuation problem should consider the optimal exercise boundary and there is no simple closed-form solution in general. Much literature thus arose on the subject of efficient approximations, and numerical methods. Barone-Adesi and Whaley [3] created a fast analytic approximation for American option prices by separating the price into a European option value plus an early exercise premium, and then approximating that premium instead of solving the full free-boundary problem exactly. Bjerksund and Stensland [4] later developed a closed-form lower-bound approximation based on feasible, but non-optimal, exercise strategies, again emphasizing computational efficiency. Ju and Zhong [5] proposed another approximation keep the same general analytical spirit of earlier fast formulas, but add a correction term to the early exercise premium so that the approximation becomes more accurate, especially for intermediate and long maturities, where older formulas could misprice more seriously. Finally, Longstaff and Schwartz [6] introduced the least-squares Monte Carlo approach, which used simulation and regression to estimate continuation values and made American option valuation much more tractable.

When combined, these works reveal an evident development of restrictive closed-form formulas towards the methods of numerical and approximation schemes with the ability to compute early exercise and more complex dynamics.

2.2 Machine Learning for option pricing and implied volatility

Machine learning is best applied to option valuation since the function relating option attributes to prices is typically nonlinear, high-dimensional and hard to define correctly in advance. An extensive survey of non-parametric option pricing indicates that machine learning has become a significant alternative to purely parametric models, particularly when it is desired that the predictive accuracy of the model is measured in realistic market conditions as opposed to adherence to a simplified structural model [7].

De Spiegeleer, Madan, Reyners and Schoutens claim that machine learning could be especially instrumental in derivatives setting, as it offers significant computational scale-ups in pricing, hedging and problem-fitting, at a minor cost of approximation error. They further observe that a lot of derivative-pricing issues precisely possess the type of recurring functional structure that can be efficiently learned by machine learning methods [2]. Equally, Culkin and Das show that deep neural networks can replicate option-pricing relationships with a very high level of precision, which proves the effectiveness of deep learning as a functional approximator in finance [8].

Ivașcu offers more applied comparative evidence, testing a variety of machine learning algorithms on option pricing and finds that machine learning methods can perform better than classical parametric benchmarks in various classes of moneyness and maturity. Flexibility is not the only appeal of machine learning in that study; it is also a way to escape the restrictive distributional and economic assumptions inherent in many traditional models. Similar studies by Gradojevic and Kukolj also report high performance of non-parametric models including random forest and extreme gradient boosting, but also highlight the value addition of explainable AI tools to understanding model behaviour [8].

This reasoning can be extended to implied volatility. Implied volatility is not a fixed value but a surface that is provided over moneyness and maturity, and the surface has consistent features across

market conditions like smiles, smirks and a varying curvature. New research on implied volatility surface forecasting demonstrates that the full surface modelling, as opposed to single points, can enhance predictive performance since volatility has both time and space structure [9]. Empirical studies of ETF options also reveal that the form of the implied volatility curve is related to market conditions, investor sentiment and risk appetite, which support the perception that IV is a complicated market phenomenon rather than a straightforward model input [7].

In general, the literature indicates that machine learning is especially useful in two cases applicable to this report: first, when the goal is to approximate a complex pricing function; and second, when the variable of interest is nonlinearly structured market, as in implied volatility.

2.3 Bid-ask spread and market microstructure

Bid-ask spreads, in contrast to theoretical option prices, are clearly market-based results, and thus they need a microstructure view. An initial understanding is provided by Glosten and Milgrom, who demonstrate that the bid-ask spread may be a mechanism of compensation to adverse selection. The existence of better-informed traders even when the specialist is risk-neutral and earns no expected profit imposes expected losses that have to be compensated by a positive spread [8]. This implies that spreads are not merely arbitrary trading frictions, but measures of information risk and liquidity provision that are economically meaningful.

Competition and trading activity are also related to the spreads in the options markets. The U.S. options markets evidence indicates that quoted and effective bid-ask spreads achieved are more likely to decrease as the competition among exchanges grows, which confirms the perspective that transaction costs depend on the market structure and liquidity conditions [7]. More generally, this literature suggests that the tighter the spreads should be between the higher trading activity and more liquid contracts and the broader the spreads of the less active and hard to hedge contracts.

The margin and capital usage is another significant consideration. CBOE margin manual indicates that uncovered short option positions may take heavy collateral and even more so than that minimum set by the exchange [10]. This translates to market makers tying up capital by the provision of liquidity. When a contract is not liquid, risky, and can not be hedged, they will tend to increase the spread to cover that capital use and risk.

Lastly, the demand specific to options can also affect the cross-sectional prices. Bollen and Whaley show that net buying pressure affects the shape of implied volatility functions [1]. That is, IV is not solely driven by true expected volatility, it is also driven by imbalances in the options market in the form of demand.

3 Data and Feature Engineering

3.1 Synthetic American option dataset

The analysis is based on a synthetic dataset of prices of American call options in the first part of the analysis. The target values were created using a binomial pricing model, which offers a controlled setting where the actual pricing model is internally consistent. This arrangement is convenient as it enables the performance of the model to be analyzed without the extra noise, frictions and quotation anomalies which is exhibited in actual markets. That is, Part 1 is intended as a benchmark task: in the case a machine learning model cannot be trained on a smooth synthetic pricing function, it is unlikely to work well in more challenging real-world tasks.

An initial look at the synthetic data indicated that the strike price and the number of binomial steps were fixed between observations, thus making them not useful cross-sectional variation to the machine learning problem. The outlier analysis by using interquartile range proposed that the primary apparent outliers were in time to maturity. These values however were a reflection of the intended economic range of the synthetic contracts and not data error, thus were kept.

It was followed by a series of transformations based on finance to add to the feature set without making it

uninterpretable. To begin with, the square root of maturity was added as a factor because in the standard option theory the squares of time are approximately proportional to the square root of uncertainty. Second, interaction terms like rT, divT and $v\sqrt{T}$ were created to reflect the cumulative impact of interest, dividend and volatility to the life of the option. Third, intrinsic value and an in-the-money indicator were added in order to reflect the immediate exercise aspect of the American call. These artificial variables aid the models in learning the structure of pricing, which would have to be learned indirectly by using the raw inputs alone.

3.2 Real-world NFLX options dataset

The second section of the research involves the real world options data that is created based on the NFLX option-chain data. Only the call options have been analyzed and the observations that had missing values on the critical variables such as the bid, ask, implied volatility, maturity, stock price, strike, volume and open interest have been pruned out before the modelling.

Since real options data have quotation noise and some anomalies, further cleaning rules were implemented. The implied volatility would only be observed between positive and economically reasonable values, spreads must be positive, volumes must be strictly positive, and moneyness was limited to eliminate extreme contracts that are not likely to be reflective of liquid and meaningful observations.

3.3 Feature engineering for IV and Spread tasks

The real-world data was then converted into features that were specific to the two prediction problems. The difference between ask and bid/midpoint quote was calculated and used as a relative measure of the bid-ask spread. This gives a scale-adjusted trading cost and liquidity. Moneyness was calculated as stock price/strike price and a log-moneyness transformation was incorporated since the raw variable was very skewed. Squared moneyness term was another term developed to describe smile type curvature of implied-volatility relationship.

The logarithms of the trading volume and open interest were used as liquidity proxies. These were added since more active contracts tend to have smaller spreads and can also include information regarding the location of the liquidity and the location of attention to the trading. Lastly, the square root of time to expiration was added as a reflection of the usual time-scaling reasoning in option theory, and to enable the models to model maturity-related effects more flexibly than using maturity alone.

The options of features varied a bit in the two real-life tasks. In the case of implied volatility prediction the emphasis was put on the variables that were correlated with the volatility surface - particularly moneyness, transformed moneyness and maturity - and liquidity variables were used as secondary predictors. To predict the spread, the feature set was the same but with a more emphasis on the predictors that were related to liquidity, as the quoted spreads are directly influenced by the trading activity, inventory risk, and market-making conditions.

4 Methodology

4.1 Modelling strategy

This study is designed empirically in three parts. Part 1 is a synthetic American call option pricing problem, in which the task is to learn a smooth pricing function that is produced under controlled conditions. Part 2 takes real-world NFLX option-chain data and divides the analysis into two distinct tasks of regression: implied volatility prediction and bid-ask spread prediction. This segregation is significant in that the three targets are different economically.

In all three tasks, the objective was to not only compare predictive accuracy among models, but also to learn what variables drive the predictions, and whether the model outputs make financial sense.

4.2 Train-test design and Stratification

A stratified sampling approach was used throughout the study. As the three tasks are all regression problems, stratification was applied by building quantile-based bins of the target variable and employing them in the train-test split. This will aid in maintaining the proportion of the target between training and testing sets, and mitigate the chance of one of them having a disproportionate number of high-value or low-value observations.

In the synthetic pricing exercise, stratified splitting was done with ten quantile bins of the American option price. Moreover, time to maturity was split and checked separately after the split since maturity is an economically significant factor and shows a large distribution in the generated data.

The same general approach was applied in the real-world tasks. To predict implied volatility, the train-test split was stratified with binned implied volatility values. To predict spread, binned spread values were used to split the prediction. This offers a more reliable point of comparison of the models than a random split that is purely random, particularly considering the relatively small size of the real-world NFLX sample after cleaning.

4.3 Model classes

Linear Regression (a benchmark model) was used, as it gives a convenient baseline as it can be readily interpreted and shows how much of the relationship can be explained by linear combinations of engineered features alone.

Support Vector Regression (SVR) model using the radial basis kernel was added since the use of the kernel methods can at once provide nonlinear relationships in a smooth way and is also comparatively strong in smaller tabular datasets. This rendered it a viable choice on the synthetic pricing problem, with the mapping involved being nonlinear but smooth.

A number of tree-based models were also employed. Only one Decision Tree was added as a poor benchmark mostly because tree models are versatile yet individual trees may be wobbly and pragmatic to overfitting. Ensemble versions with more power were subsequently applied: Random Forest, Gradient Boosting, XGBoost, and LightGBM. These models are particularly well adapted to structured financial data since they are able to include nonlinearities and interactions between features without strong distributional assumptions. According to prior literature, these methods also work well in option-pricing tasks, and tend to compete effectively with neural networks in tabular data.

Lastly, a feedforward neural network was developed with the use of Keras. Dense layers were selected due to the tabular nature of all datasets as opposed to sequence. The neural network in the synthetic pricing exercise is used as a non-rigid approximator of the option-pricing surface. It provides further test of whether more model flexibility enhances prediction at the real-world tasks as compared to tree-based methods.

4.4 Pipelines, scaling, and reproducibility

In the synthetic pricing task, traditional models were handled with pipelines in all cases where feature scaling was necessary. Specifically, a Pipeline that included Linear Regression and SVR with a StandardScaler was used. This made sure that scaling was only applied to the training folds in cross-validation which prevented data leakage. The tree-based models were not scaled since they are mostly not sensitive to monotonic rescaling of the features.

In the real world tasks, scale sensitive models were once more trained on standardized inputs. Linear Regression, SVR and the neural network processed scaled feature matrices, whereas the Random Forest, Gradient Boosting, XGBoost and LightGBM were trained on the original features.

Reproducibility was taken into consideration by correcting a common random seed during the analysis.

4.5 Tuning and Optimization

Linear Regression was kept as an untuned baseline. For all other traditional models, either cross-validation or randomized search was used. Instead of the exhaustive grid search, randomized search was used due to the trade-off between predictive performance and computing costs of the search. The search of SVR was done lightly since the radial basis kernel approaches may be computationally intensive with large samples.

In Part 1, the neural network has been optimized less intensively based on architecture selection and early stopping as opposed to a large hyperparameter search. The chosen architecture consisted of two hidden layers that were replaced with ReLU activations, Adam optimization, and early stopping via validation loss. This was adequate since the aim was not to comprehensively optimize the deep-learning performance but to evaluate whether a reasonably-designed feedforward network could learn the pricing function with sufficient precision.

The main comparisons in the real-world tasks were carried out with fixed yet reasonable specifications. This choice was due to the reduced size of the dataset and the practical necessity to retain the modelling workflow to an acceptable size. Further testing was subsequently done on XGBoost in the spread model to determine whether more regularization or other parameter selection would yield better generalization.

4.6 Evaluation metrics

Four regression measures were used to determine model performance:

- **Mean Squared Error (MSE)**
- **Root Mean Squared Error (RMSE)**
- **Mean Absolute Error (MAE)**
- **R-squared (R^2)**

This combination was selected since each measure offers a slightly different information. MSE and RMSE resolve the errors that are bigger more severely and are thus applicable in the event that the big errors are of great disfavor. MAE is more easily interpreted since it shows the average absolute deviation of the true value. R^2 is a scale free measure of goodness of fit and measures the extent to which the variation in the target is explained by the model.

In the case of financial realism, a prediction was clipped at zero and the negative values would not be reported in a manner that would not make economic sense.

4.7 Interpretation and Diagnostic tools

To get the best synthetic pricing model, SHAP was used via a model-agnostic KernelExplainer since the best model was a neural network. This enabled the contribution of features worldwide to be evaluated despite the model not being interpretable itself. Further, the grouping of the test sample into out of the money, at the money, and in the money regions was also carried out in order to determine whether the quality of prediction in different moneyness regimes differed.

On the best tree-based model, both permutation importance and SHAP were applied to the implied-volatility task. Permutation importance offers a simple metric on how much predictive performance would be damaged were any single feature to be shuffled, whereas SHAP offers a more detailed breakdown of the contribution of features.

On the spread task, SHAP was again applied to the highest-performing tree-based model with a view of determining whether the most critical drivers reflected by machine learning were the same as the financial intuition about moneyness, maturity and liquidity.

5 Empirical Results

5.1 Part 1: Synthetic American option pricing

Table 1: Part 1 synthetic pricing: test-set performance by model

Model	**Test MSE**	**Test RMSE**	**Test MAE**	**R^2**
Keras Neural Network	0.005648	0.075150	0.049992	0.999961
SVR	0.016116	0.126949	0.050044	0.999890
LightGBM	0.219305	0.468300	0.270575	0.998504
XGBoost	0.237977	0.487829	0.327944	0.998377
Random Forest	0.747715	0.864705	0.433943	0.994900
Linear Regression	1.998825	1.413798	1.010036	0.986366
Decision Tree	2.377586	1.541942	0.886727	0.983782

Synthetic pricing generated quite high results in most of the models to be expected in a controlled environment where the target is created by a stable pricing mechanism. The Keras neural network had a superior overall performance, with the lowest MSE, RMSE, and MAE with the highest R^2. SVR came in second and provided a very close approximation to the MAE, but the RMSE was larger, indicating that the neural network was better at dealing with large errors in pricing.

LightGBM and XGBoost were the best tree-based models, whereas the single Decision Tree was the worst model. This corresponds to the nature of the problem, a synthetic option-pricing function is nonlinear but smooth so those models which are able to approximate smooth nonlinear surfaces are the best performers. Linear Regression was also quite successful and this indicates that much of the option-pricing structure was already represented by the engineered features.

Table 2: Pricing performance across moneyness buckets (S/K)

Bucket	**Count**	**MSE**	**RMSE**	**MAE**	**RMSE / Mean True Price (%)**
OTM	2884	0.004	0.063	0.041	1.014
ATM	424	0.007	0.084	0.058	0.671
ITM	2927	0.007	0.084	0.058	0.355

Further bucket analysis by OTM, ATM and ITM regions indicated that the model was still precise through the cross-section though there were relative errors that were slightly high in the low-premium OTM.

5.2 Part 2A: Implied volatility prediction

The task that implied volatility was more difficult since the target captures actual market behaviour as opposed to a theoretical pricing rule. In this case, Random Forest was the most efficient, then came XGBoost and lastly, the Linear Regression was the least efficient model. This ranking is intuitive in that implied volatility is highly nonlinear with respect to strike and maturity, and tree-based ensembles are well adapted to represent such a structure.

Table 3: Implied volatility prediction: test-set performance by model

Model	**MSE**	**RMSE**	**MAE**	$\mathbf{R^2}$
Random Forest	0.010892	0.104364	0.040203	0.995317
XGBoost	0.015041	0.122643	0.050565	0.993533
LightGBM	0.029124	0.170659	0.071699	0.987477
Neural Network	0.031577	0.177700	0.094396	0.986423
Linear Regression	0.474694	0.688980	0.401180	0.795897

The methods had a high degree of consistency in results of feature-importance. The most important predictors were moneyness, log-moneyness, and squared moneyness, with variables related to maturity being secondary, and liquidity variables having limited explanatory power. This is consistent with the intuitive view of volatility smiles and skews: IV is predominantly determined by the position of the option in relation to the strike and the curvature of the volatility surface.

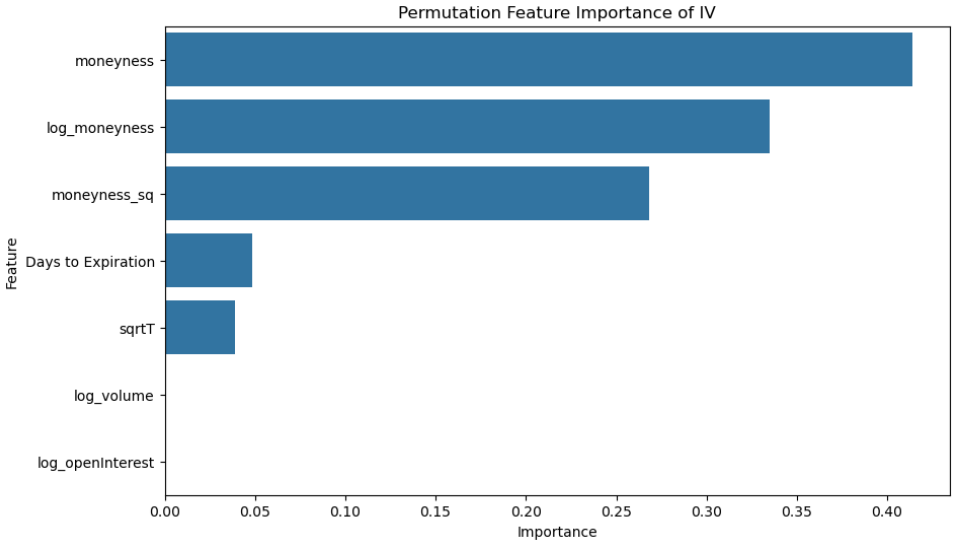

Figure 1: Permutation feature importance for the best implied volatility model

One more visual observation was that the implied-volatility surface was steeper towards short-dated options and flatter towards long-dated ones. This has an economic significance. Short-dated contracts are more prone to local strike specific risk and short run market pressure whereas the longer dated contracts are more averaged expectations of uncertainty and thus the smoother surface.

Results of interpretability analysis indicated that log(openInterest) had very little predictive value and therefore the models were rerun without it. There was no major or minor change in performance, which implies that this variable was redundant or non-informative in the IV specification.

5.3 Part 2B: Bid-ask spread prediction

Table 4: Bid-ask spread prediction: test-set performance by model

Model	MSE	RMSE	MAE	R^2
XGBoost	0.045024	0.212188	0.091782	0.893592
LightGBM	0.047474	0.217886	0.099249	0.887800
Random Forest	0.049440	0.222351	0.096963	0.883155
Gradient Boosting	0.054172	0.232748	0.103469	0.871972
Neural Network	0.070401	0.265333	0.139660	0.833615
SVR	0.070838	0.266153	0.151668	0.832584
Linear Regression	0.141305	0.375905	0.271475	0.666043

The spread task was less accurate at predicting than the pricing and IV tasks, which is not surprising since quoted spreads are an indicator of market frictions, liquidity state, and trading behaviour instead of a consistent theoretical mapping. XGBoost on all models was the best, and LightGBM and Random Forest were close to it. Linear Regression was significantly weaker, which implies that the formation of spreads is a result of nonlinear interactions.

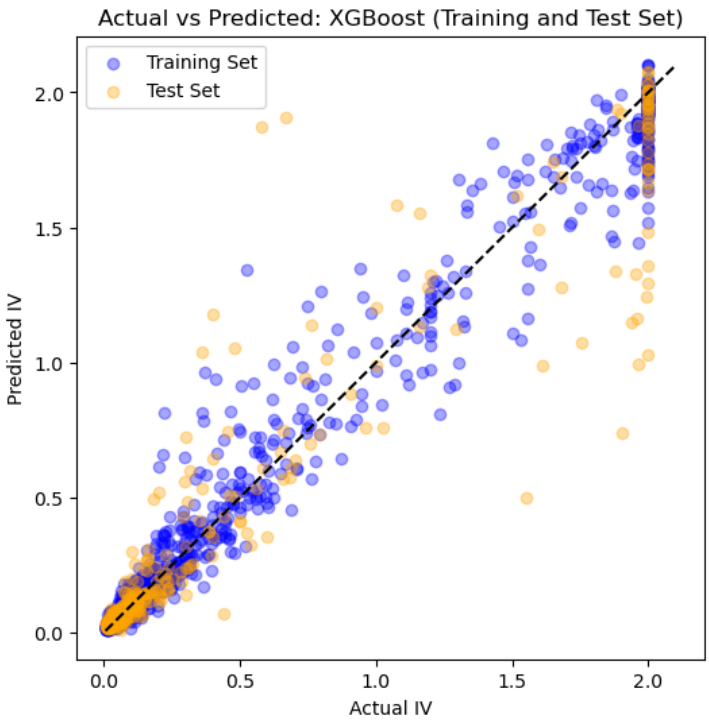

Figure 2: Training and Testing fitting on the best spread model

The comparison of training-versus-tests indicated that there was some overfitting in the best XGBoost model. Nevertheless, despite this shortcoming, its performance out-of-sample was the best among the candidate models hence it was kept as the top specification.

The SHAP analysis revealed that the most significant predictor of the spread was moneyness, then moneyness squared, maturity-related variables and finally the liquidity proxies. This finding makes financial sense. Contracts that were lower-moneyness or further out-of-the-money were more likely to be wider spread, whereas more in-the-money contracts were more likely to be narrow. This is in line with reduced liquidity and increased difficulty in hedging contracts that are less actively traded. The wider spreads were also related to shorter-dated options, which conforms to the notion that the gamma risk

increases as one approaches expiration. Lastly, the increased volume and open interest were correlated with smaller spreads, which is in line with the perception that increased liquidity decreases transaction costs and inventory risk.

6 Discussion

The three empirical exercises lead to a number of general conclusions.

First, the performance of models is very sensitive to the nature of the target variable. In the synthetic pricing task, the target was created based on a stable functional process, hence the highly flexible models including neural networks and SVR worked very well. Conversely, the real tasks were more noisy in the sense that the targets were market results as opposed to the theoretical values. That is why the predictive accuracy has decreased between Part 1 and Part 2.

Second, the most effective model type was different in tasks. The neural network was robust with synthetic pricing, in which smooth functional approximation was to be pursued. In comparison, tree-based ensemble techniques dominated the IV and spread problems. This trend is in line with the larger literature indicating that neural networks are particularly effective at flexible function approximation, and tree ensembles are especially effective in tabular structured financial data.

Third, the interpretability results show that the machine learning models were not only accurate, but also economically sensible. In the synthetic pricing task, the neural network identified stock price, time scaling, and interactions related to volatility as some of the main drivers. The dominance of moneyness and smile related variables in the IV task was precisely what the option theory would forecast. Moneyness and maturity were again the strongest predictors in the spread task, which was complemented by liquidity measures. This reinforces the notion that the models were capturing significant financial structure as opposed to random statistical noise.

7 Limitation

First, the real-world analysis is conducted on a one-cross-sectional snapshot of NFLX call options, as opposed to a time-long panel. Consequently, the models do not represent the entire time-series behavior.

Second, the real world data lacks more detailed information on the microstructure like updates on quotes, depth of a bid and ask or direction of a trade and imbalance of an order book.

Third, Part 2 deliberately restricted hyperparameter tuning compared to Part 1 due to computational cost and scope of the project. Perhaps, more extensive tuning, particularly of the boosting models and the neural network, would slightly enhance performance.

Fourth, the sample in the real world is limited to NFLX call options alone, and therefore, results cannot be considered to be generalizable to all underlying and all types of options.

8 Conclusion

This report discussed machine learning usage in three related option-market contexts: synthetic American option pricing, implied volatility prediction, and bid-ask spread prediction.

The findings demonstrate that machine learning has the capability to estimate the relationships in option pricing to a great extent especially in controlled environments. The neural network in the synthetic pricing task also demonstrated close to perfect predictive accuracy, validating that nonlinear models with flexibility can be used to learn complex pricing surfaces with high accuracy. Tree-based ensemble methods were found to be the most effective in the real-world tasks. Implied volatility was better predicted by Random Forest, and the best spread predictions presented by XGBoost.

The interpretability analysis revealed also that the models were economically consistent. In the IV task, the variables of moneyness and smile related dominated. Moneyness, maturity and liquidity measures

were the dominant measures in the spread task. The results are in line with the option-pricing theory and market microstructure intuition.

On the whole, this evidence indicates that machine learning can be used as a derivative-market analysis tool, not to substitute financial theory, but rather as a versatile complement to it. Machine learning has the potential to deliver high predictive accuracy and economically valuable insights when backed by sound feature engineering and prudent interpretation.

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